



UNIVERSITY OF PERADENIYA
DEPARTMENT OF STATISTICS AND COMPUTER SCIENCE
END SEMESTER EXAMINATION, SEMESTER II – JANUARY, 2022
ST/MT102 – INTRODUCTION TO PROBABILITY THEORY – PART I

Time Allowed: **TWO Hours**

Answer **All** Questions.

✓ 1.

- a). Prove the following identity. [20 marks]

$$\binom{n}{r} \binom{r}{k} = \binom{n}{k} \binom{n-k}{r-k}$$

- b). A group of 5 men and 4 women are to be faced for a photograph in a row.
- i) How many different arrangements are possible if three are sitting and the rest are standing at the back? [15 marks]
 - ii) How many different arrangements are possible if all women are sitting and all men are standing at the back? [15 marks]
 - iii) If all 9 of them are sitting for the photograph in a row, what is the probability that all women sit together? [20 marks]
- c). Suppose that 5% of men and 0.25% of women are colour-blind. If a person is chosen at random and that person is colour-blind, what is the probability that the person is a male? [Assume males and females are to be in equal numbers]. [30 marks]

✓ 2.

The probability density function (pdf) of a random variable X is given by

$$f_X(x) = \begin{cases} c(x-1)^2 & ; \quad 0 < x < 2 \\ 0 & ; \quad o/w \end{cases}$$

- a). Find the value of c such that $f(x)$ is a legitimate pdf. [15 marks]
- b). Derive the Cumulative Density Function (CDF) of X . [20 marks]
- c). Calculate $P(X > 1)$. [20 marks]
- d). Calculate $E(X)$. [20 marks]
- e). Find $P(X > \frac{3}{2} | X > 1)$. [25 marks]

[P.T.O.]

3.

a). Let X be a continuous random variable where the moment generating function, $M_x(t)$ exists.

i) For constants a and b , show that in usual notation, [25 marks]

$$M_{\frac{(x+a)}{b}}(t) = e^{\left(\frac{a}{b}\right)t} M_x(t/b)$$

ii) If the random variable X has the continuous uniform distribution within the interval (a, b) , show that the moment generating function (MGF) of X is [15 marks]

$$M_x(t) = \frac{e^{bt} - e^{at}}{t(b - a)}$$

iii) Derive an expression for the r^{th} moment about the origin and use it to find the mean of X . [15 marks]

b). Suppose $X \sim \text{Exp}(\lambda = 2)$.

i) Find a lower bound for the probability of X within two standard deviations of mean.

State the inequality you used to obtained the answer. [25 marks]

ii) Calculate the exact probability for part (i). [20 marks]

4.

a). If $X \sim \text{Bin}(n, \theta)$ and $Y \sim \text{Bin}(n, 1 - \theta)$, show that $P(X = x) = P(Y = n - x)$. [20 marks]

b). When a fair six sided dice is rolled 8 times, what is the probability that outcome 5 occurs

i) exactly 3 times? [10 marks]

ii) for the first time in the 4th roll? [10 marks]

iii) for the 3rd time in the 6th roll? [10 marks]

b). The weight loss (in kg) of participants in a certain diet programme is normally distributed with mean 5 and standard deviation 2.

i) What percentage of participants will end up with more than 7kg of weight loss? [15 marks]

ii) Calculate the 90th percentile for weight loss. [15 marks]

iii) What percentage of participants will fall within one standard deviation of the mean weight loss? [20 marks]

Distribution	Density Function
Binomial (n, θ)	$P(X = x) = \binom{n}{x} \theta^x (1 - \theta)^{n-x}; x = 0, 1, 2, \dots, n$
Negative Binomial (k, θ)	$P(X = x) = \binom{x-1}{k-1} \theta^k (1 - \theta)^{x-k}; x = k, k+1, k+2, \dots$
Geometric (θ)	$P(X = x) = \theta (1 - \theta)^{x-1}; x = 1, 2, \dots$
Exponential (λ)	$P(X = x) = \frac{1}{\lambda} e^{-\frac{x}{\lambda}}; x > 0$ where $E(X) = \lambda, \text{Var}(X) = \lambda^2$
Uniform(a, b)	$f(x) = \frac{1}{b-a}; a \leq x \leq b$